Hamdan bin Rashid Al Maktoum Foundation for Medical and Educational Sciences (HF 2026)
1558-BRH Best Research in Healthcare

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DEVELOPMENT OF AN ADAPTIVE SAMPLING-BASED LONG-READ SEQUENCING ASSAY FOR PREMARITAL GENETIC SCREENING

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Curriculum Vitae of First Author

DOCX	CV 2026.docx (48 KiB download)

Curriculum Vitae of PI

PDF	C.V - AHMAD ABOU TAYOUN_M... (310 KiB download)

Additional Curriculum Vitae	Yes

Additional Curriculum Vitae #2

PDF	Sathishkumar_Ramaswamy_CV... (85 KiB download)

No Objection Letter

PDF	BRH Declaration-2.pdf (476 KiB download)

First Page

DOCX	Title Page_HA.docx (29 KiB download)

Research Paper

DOCX	Hamdan Award .docx (4.3 MiB download)

Supplementary Materials

XLSX	Supplementary Folder HA.x... (22 KiB download)

Has the submitted research received any awards?	No

Layperson's Summary

> Standard genetic tests used in premarital screening read DNA in very short fragments, like trying to solve a jigsaw puzzle using only tiny pieces. While this works for most of the genome, it consistently fails in regions where genes look almost identical to one another, such as the genes responsible for thalassemia, spinal

muscular atrophy, and congenital adrenal hyperplasia. These are precisely the conditions most relevant to high risk populations, meaning some carriers are missed entirely.

This project developed a new type of genetic testing using a technology calling Oxford Nanopore long-read sequencing, which reads much longer stretches of DNA at once, like assembling a puzzle with larger, easier to place pieces. Crucially, it was combined with a technique called adaptive sampling, where the sequencing machine makes real-time decisions about which DNA fragments to read and which to discard. This allows it to focus exclusively on the 836 genes relevant to inherited disease, without needing any physical chemical enrichment steps.

The assay was built, optimized, and tested on 75 individuals with previously confirmed genetic variants spanning multiple disease categories. It successfully detected all variant types, including small mutations, structural rearrangements, and large deletions with 100% accuracy. Notably, it resolved complex gene regions that routinely defeat standard tests, all within a single unified workflow that eliminates the need for several separate follow-up tests currently required in clinical practice.

The results is a faster, more accurate, and more comprehensive genetic screening tool that can be realistically scaled to a population level. By catching variants that existing methods miss, this assay has the potential to meaningfully reduce the rate of missed at-risk couples in premarital screening programmed with direct implications for disease prevention and family health planning.

Relevance - 10%

/ 5.00

5 being the highest and 1 as the lowest. * The research topic is highly relevant to the respective field. * The research topic addresses a significant gap or problem in healthcare within its respective field. * The research demonstrates potential for practical application or impact on current issues.

Study design - 25%

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Results and findings - 25%

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Impact potential - 20%

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5 being the highest and 1 as the lowest. * It has the potential to make a significant impact on the field. * Findings contribute to policy changes or future practice. * Findings may benefit patient care or public health.

Presentation and clarity - 20%

/ 5.00

5 being the highest and 1 as the lowest. * The paper is well organized, clear, and concise. * The visual aids enhance the understanding of the research. * Figures and tables are clear and effective at presenting data. * The sources are relevant and sufficiently recent. * The writing style is professional.

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